Animesh Kumar

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LinkedIn
GitHub
Portfolio

Professional Summary

AI Engineer (MS, GPA 4.0) with experience building end-to-end recommendation, retrieval, and agentic AI systems. Skilled in learning-to-rank models, transformer-based architectures, and production AI deployment using PyTorch, Spark, and AWS. Published author in deep learning and computer vision (ACCAI 2024, INFUS 2024), focused on optimizing user engagement and personalization through data-driven ML solutions.

Education

Arizona State University, MS in Computer Science – Tempe, AZ — May 2026

- GPA: 4.0/4.0

MIT World Peace University, BTech in ECE – Pune, India — July 2024

- GPA: 3.9/4.0

Technical Skills

Languages & Frameworks: Python, MongoDB, PyTorch, TensorFlow, Keras, HuggingFace Transformers, XGBoost, scikit-learn, SHAP

ML & AI: Recommender Systems, NLP, Computer Vision, CNNs, RNNs, Transformers, LLMs, RAG, Agentic AI, CrewAI, AutoGen, Information Retrieval, Model Evaluation, Prompt Engineering, Vector Databases

Data & MLOps: Apache Spark, Pandas, NumPy, Neo4j, FastAPI, Docker, Kubernetes, AWS, GCP, Cloud Run, Vertex AI, A/B Testing, CI/CD, Git, NoSQL, Distributed Systems, Model Monitoring, Unit Testing

Experience

IT Specialist / Data Scientist, Trinity Tree – Remote — June 2026 – Present

- Co-built a production LangGraph pipeline on Cloud Run, rebuilding a local-only no-code prototype to draft posts for 3 platforms via a cross-provider writer–critic loop (GPT-5.5, Gemini) that checks claims against the practice's website
- Owned the approval layer (durable pause/resume, 24h reminder, 48h expiry) and fixed a flaw where email security scanners silently executed review decisions, by splitting each decision into a read-only confirmation page and a POST action

Software Engineering Intern, SAMSAN Technologies – Pune, India — July – December 2023

- Developed a CNN-based smoking detection model (VGG16, transfer learning) achieving 88% accuracy at 15 FPS, then integrated it into a real-time driver monitoring pipeline with facial landmark-based drowsiness detection, applying probability thresholding to minimize false alarms

Projects

MatchDay: Agentic AI Assistant with MCP Integration and RAG Pipeline — Summer 2026

- Built conversational football agent orchestrating Gemini 3.1 Pro through a custom Python tool-calling loop with 21 tools spanning grounded web search, MongoDB MCP for user state, and Discovery Engine RAG; deployed on Cloud Run
- Raised RAG top-2 hit rate from 58% to 92% by building a 28-query labeled eval harness and diagnosing a relevance threshold that discarded correct results

Ticket Router: Fine-Tuned LLM Pipeline for Support Triage — Spring 2026

- Fine-tuned DistilBERT classifier and Llama 3.2 3B with QLoRA (4-bit, r=16) on ASU A100 supercomputer via SLURM, achieving 99.9% accuracy on 4K held-out samples across 4 categories and 4 priority levels
- Chained classifier and LoRA adapter into end-to-end inference pipeline; deployed on HuggingFace Spaces with Gradio frontend, models hosted publicly on HuggingFace Hub

SkillBridge: Multi-Agent Skill Gap Analyzer with CrewAI — Summer 2026

- Built two-crew pipeline using crewAI Flows orchestrating four specialized agents (JD analyzer, resume analyzer, course researcher, course curator) with Llama 3.3 70B and 3.1 8B via Groq, connected by Pydantic-validated state
- Implemented custom LeanSerperTool reducing token usage by 80% and monkeypatched crewAI cache injection bug breaking Groq compatibility; deployed on HuggingFace Spaces with Gradio frontend

DebateClub: Multi-Agent Debate System with AutoGen — Summer 2026

- Built structured debate system using AutoGen 0.2.35 with four agents (debater, scorer, verdict, final judge) orchestrated over Gemini 2.5 Flash via Vertex AI OpenAI-compatible endpoint
- Implemented Pydantic v2 validation with single-retry self-correction on malformed JSON and deterministic arithmetic tally to prevent LLM score tampering; deployed on HuggingFace Spaces